

JONS JACOB BERZELIUS

Swedish professor Jöns Jacob Berzelius developed the modern system of letters for atomic symbols.



In 1807, Berzelius (1779–1848) became professor of chemistry at Sweden's Karolinska Institute. He researched catalysts and isolated thorium, cerium, and selenium. A proponent of Dalton's atomic theory, he attempted to work out elements' atomic weights more accurately. Dalton had compared elements to hydrogen; Berzelius used oxygen as his standard. In his new system, published in 1814, he symbolized elements with the first letter of their Latin name (such as C for *carbo*, the Latin for carbon). He later added, as subscript, the number of atoms in a compound (as in H_2O).

LISTED THE RELATIVE WEIGHTS OF 20 ELEMENTS



MADE OVER
200,000
OBSERVATIONS
IN HIS
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observation that the pressure of a volume of gas in a container depends on its temperature, and that given sufficient pressure and low temperature, all gases become liquid.

All gases exert pressure on the walls of the container that holds them. Dalton observed that even if there were several gases in a container, each one exerted the same pressure as it would if it were alone in the container. The total pressure of the combined gases was equal to the sum of these partial pressures. This became known as Dalton's law of partial pressures and remains a fundamental principle in modern chemistry.

Atomic theory

The idea of atoms (tiny indivisible particles) dated back to the ancient Greeks, but it was assumed that atoms were identical in every element. However, Dalton had noticed from his experiments that oxygen absorbed less water vapor than nitrogen. He wrote of this: "I am nearly persuaded that the circumstance depends on the weight and number of the ultimate particles of

the several gases." In other words, he wondered if oxygen and nitrogen had atoms of different weights.

If this was the case for oxygen and nitrogen, Dalton reasoned, it must be so for every element. He ran a series of experiments comparing the known elements with hydrogen (the lightest of the elements) and assigned them each a number that represented their relative atomic weights. He went on to create the first atomic table featuring all the elements known at the time, each with a pictorial symbol. The weights he calculated were not always accurate because he did not realize that atoms of the same element might combine. For example, he assumed that oxygen was a single atom rather than a molecule with two atoms. However, later scientists were able to calculate the weights more accurately, and Dalton's principles form the foundation of modern chemistry.

In his book *A New System of Chemical Philosophy* (1808), Dalton set out his atomic theory. He said atoms were indivisible and could not be created or destroyed; each element had atoms of a unique mass (weight); and atoms of different elements combined in chemical reactions to form compounds in simple whole-number ratios. The atomic theory was quickly accepted, and Dalton was elected to the Royal Society in 1822. A few years later, Jöns Jacob Berzelius refined Dalton's atomic table using letters rather than pictorial symbols, but Dalton preferred his own symbols and did not adopt the new system.

